Paleomagnetic and Geological Implications of Magnetic Properties of the Triassic Diabase of Southeastern Pennsylvania

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Abstract. Preliminary measurements of magnetic properties of Triassic diabase in southeastern Pennsylvania imply a Triassic geomagnetic pole in central Siberia, at 62°N, 105°E. This pole is virtually identical to that obtained from paleomagnetic studies of rocks of similar age in New Jersey and Arizona. A North American Triassic pole, the mean of six independent determinations, is located at 62°N, 101°E. Dispersion in directions of magnetization, examined to estimate the intensity of Triassic geomagnetic secular variation, is less than half that predicted by Creer (1962) from an analysis of the present field, which may indicate an episode of relative geomagnetic 'quiet' during the Late Triassic. These magnetic measurements also define certain linear distributions of site-mean directions of magnetization, probably caused by differential rotations of diabase units about the regional geologic strike. A preliminary tabulation of magnetic properties suggests several correlations with lithology, including the observation that the strongest, but least stable, magnetic moment is associated with the late-stage granophyric diabase facies. The effective average ratio of remanent to induced magnetization for these intrusions probably falls in the range 1.60 to 1.89.

Introduction. During 1963 a sampling program was initiated in diabase intrustions of southeastern Pennsylvania in an effort to determine magnetic and other physical properties for use in an integrated geophysical study of the Triassic basin. To date, sufficient magnetic measurements have been made to permit the calculation of a Triassic virtual geomagnetic pole. Certain distributions of directions of magnetization also have emerged which seem to have some bearing on the structure of the diabase bodies. These topics are discussed in the present paper, and a preliminary tabulation of some magnetic properties and specific gravity of Pennsylvania diabase is presented.

Geological summary. Rocks of Triassic age in Pennsylvania are exposed in a belt approximately 225 km long and 6 to 50 km wide which trends approximately east-northeast across the southeastern corner of the state. This belt of exposures is part of a longer outcrop belt, the Gettysburg-Newark basin, which is the largest of the Triassic basins of the eastern United States. To date, sampling has been confined to the southwestern three-quarters of the Pennsylvania part of this belt, from the Maryland border to a point several miles southeast of Birdsboro (see Figure 1).

Triassic sedimentary rocks in parts of this belt may have an aggregate thickness of more than 6000 meters [*Stose,* 1932, p. 55]. They consist for the most part of continental clastic deposits, ranging from shale to conglomerate, much of which is of typical 'redbed' lithology. These rocks have been assigned to the Upper Triassic [*Willard et al.,* 1959, p. 107] on the basis of rather meager faunal evidence. Contained within these Triassic rocks is a considerable quantity of diabase, occurring as sills, dikes, and irregular cross-cutting bodies which vary in thickness from less than 1 to nearly 1000 meters. Contact metamorphism around these intrusions has produced aureoles of dense, dark hornfels. A few basalt flows are recognized in the Triassic of Pennsylvania [*Stose,* 1932, p. 76; *Williard et al.,* 1959, p. 152], and in New Jersey a large amount of extrusive material is found interbedded with some of the younger Triassic sedimentary rocks of the Gettysburg-Newark basin. This igneous activity is assigned by many authors to late stages of Triassic sedimentation [*Stose,* 1932, pp. 92–93; *Willard et al.,* 1959, pp. 222–225].

The prevailing dip in the Triassic outcrop belt of Pennsylvania is toward the north or northwest, although locally this relationship is

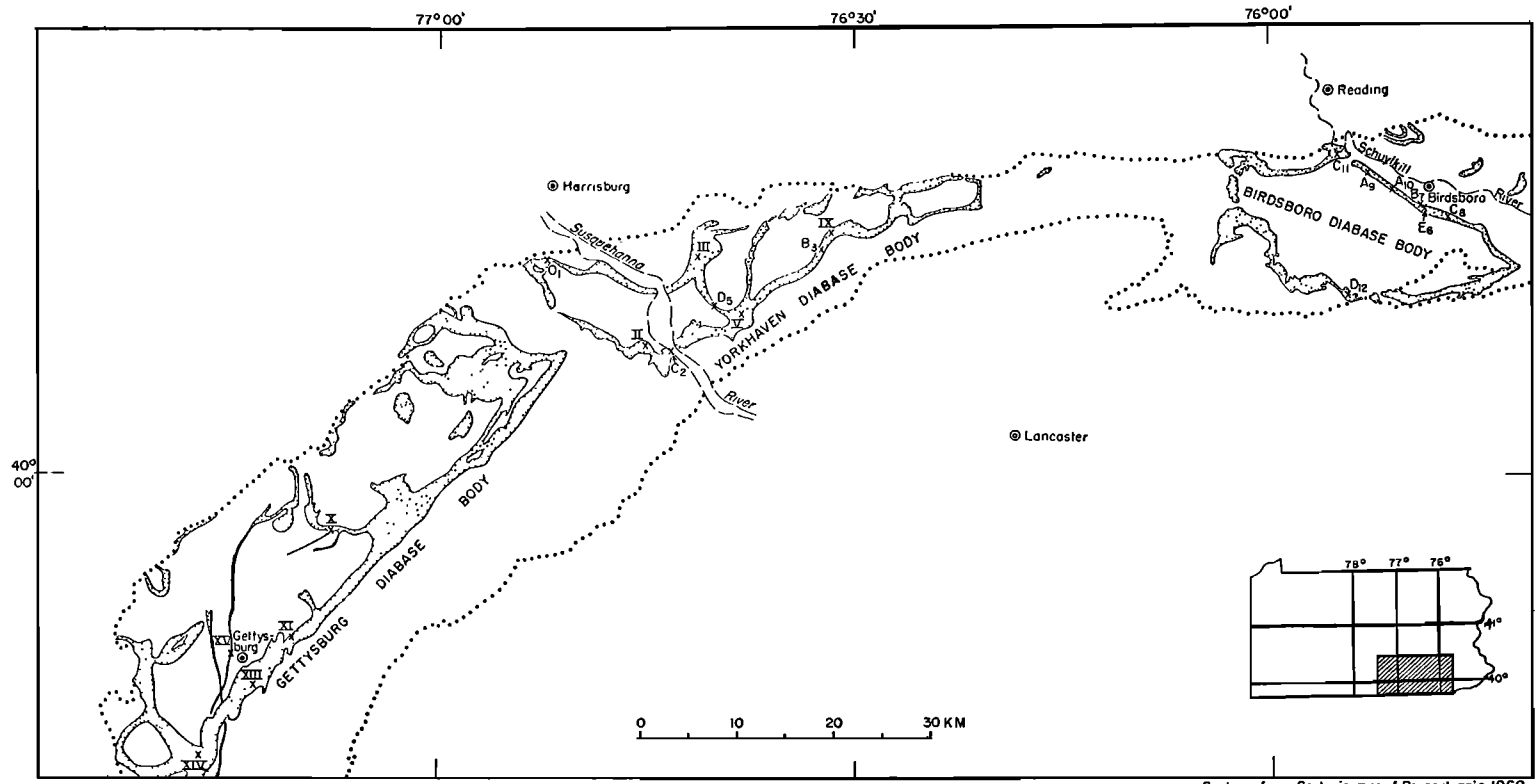


Fig. 1. Index and locality map. Sampling sites indicated by crosses and numbered as in Table 1. Letters with numeral subscripts refer to traverses; roman numerals designate isolated sites. Stipple pattern and black areas represent diabase; heavy dots indicate boundary of the Triassic basin. Geology from geologic map of Pennsylvania, 1960.

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modified by transverse warping and faulting. Most explanations for this homoclinal structure involve gradual subsidence of the Triassic basin during sedimentation, culminating in block faulting immediately after intrusion of the diabase [*Stose*, 1949; *Willard et al.*, 1959, pp. 222–225]. An interesting alternative has been advanced by *Sanders* [1963], who proposed a four-stage evolution for the Triassic basins of eastern North America, originating in a rift-valley graben. He postulated the following episodes: (1) initial graben subsidence, sedimentation, and basaltic igneous activity; (2) longitudinal arching near the axis of the graben, accompanied by uplife and erosion, imparting a homoclinal dip to the northwestern limb of the arched layers, which have been preserved as the Gettysburg-Newark basin; (3) a second period of graben subsidence, for which no record of sedimentation has been preserved, accompanied by the development of transverse folds; (4) a final stage of faulting, some of which is strike-slip with displacements of as much as 19 kilometers, by the development of longitudinal folds, and by a second generation of igneous activity resulting in the injection of porphyritic diabase dikes. Sanders' hypothesis is particularly interesting from the standpoint of the paleomagnetic data, as it proposes the existence of two periods of igneous activity separated by a period of deformation.

A striking feature of Appalachian structure in Pennslyvania is a broadly arcuate pattern defined by the trend of Paleozoic fold axes. This curvature is mirrored by the trend of the Triassic outcrop belt (see Figure 1). *Drake and Woodward* [1963] have attributed this arcuate pattern to northwestward movement of a crystalline block located near the city of Baltimore. As a consequence of this movement the Appalachian structures are believed to have been bent into their present pattern, and a major right-lateral strike-slip fault developed. As pictured by *Drake and Woodward* [1963, p. 52] this fault strikes nearly east-west, passing through the area of Figure 1 approximately at the latitude of the northern edge of the Triassic basin near Birdsboro. Inasmuch as this postulated lateral bending should produce measurable rotations of the directions of magnetization of the rocks, it is amenable to testing by paleomagnetic studies.

Field procedure. To date, 491 oriented samples have been collected from 98 sites in diabase or baked sedimentary rock. In general, five samples, either hand specimens or cores drilled in situ with a portable diamond drill, were collected at each site. Samples were oriented by means of a Brunton compass with an accuracy of 3° or better. All sites were located in artificial exposures (road cuts, railroad excavations, quarries).

Wherever exposures permitted, sampling sites were spaced at regular intervals across the strike of the diabase body. This procedure was followed in order to permit study of variations in magnetic properties caused by difference in lithology and cooling rate and as a means of detecting shifts in direction of magnetization that might be attributed to secular variation of the Triassic geomagnetic field. This phase of the investigation is still in progress. Where conditions of exposure precluded sampling traverses, isolated sites were selected which included the widest possible range of outcrop and lithology.

Laboratory procedure. Of the original 98 sites, 20 were selected for examination as part of a preliminary survey. These 20 sites were so chosen as to include the widest possible geographic area and range of lithology. One laboratory specimen was selected for measurement from each sample from a given site. Directions and intensities of remanent magnetization were measured on a motor-driven spinner magnetometer, designed and described by *Doell and Cox* [1965]. The estimated accuracy of directional measurements is 2° or less [*Doell and Cox*, 1963, p. 1998]. Magnetic susceptibility was measured on an inductance bridge, described by *Anderson* [1961].

Measurements of natural remanent magnetization showed that samples from at least 15 of the 20 sites possess significant amounts of unstable magnetization. Most of these sites show random scatter; however, in a few the directions of magnetization are streaked toward the present direction of the earth's field, which suggests viscous magnetization. All samples that showed the effects of an unstable magnetic moment responded well to alternating-field demagnetization.

A peak field of 150 oe was selected as the optimum for alternating-field demagnetization [*Irving et al.*, 1961]. This same field strength

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TABLE 1. Mean Directions of Magnetization and Precision Parameters for 20 Sites in Triassic Diabase

		Not Corrected for Tilt of Beds						Attitude of Host Strata		Corrected for Tilt of Beds		
S	N	D_0	I_0	K_0	D_{150}	I_{150}	K_{150}	Azimuth	Dip	D_{150}	I_{150}	α_{95}
					Birdsboro Diabase Body							
E_6	5	18.5	−16.5	2.0	12.0	38.5	53.6	75	25N	7.0	15.5	10.5
B_7	5	6.5	52.0	12.1	4.0	51.5	182.8	75	25N	358.0	27.0	5.5
C_8	5	356.0	29.0	21.1	7.0	40.5	282.8	80	25N	3.5	16.0	4.5
A_9	5	337.0	27.0	1.9	359.0	57.0	30.0	35	20W	342.0	42.5	14.0
A_{10}	4	343.5	48.0	21.5	6.5	46.0	60.2	40	20W	354.0	33.0	12.0
C_{11}	4	1.0	54.0	9.7	359.0	57.5	100.3	90	40N	359.5	17.5	9.0
D_{12}	5	32.0	62.5	6.1	1.0	49.5	164.9	70	30N	355.0	21.0	6.0
					Gettysburg Diabase Body							
X	4	13.0	38.5	28.5	31.0	44.5	28.4	50	25N	13 0	32.0	17.5
XI	5	12.0	37.5	451.2	10.5	37.0	335.0	50	20N	2.0	23.0	4.0
XIII	4	16.0	36.5	385.8	14.5	36.5	1043.2	25	20W	2.0	30.5	3.0
XIV	5	2.5	27.5	26.3	7.5	32.0	61.0	40	25W	359.0	17.0	10.0
XV	5	31.5	26.0	12.6	16.0	38.5	103.1	20	20W	2.0	34.5	7.5
					Yorkhaven Diabase Body							
O_1	5	353.0	34.0	7.0	11.5	47.5	177.0	80	45N	5.0	4.5	6.0
C_2	5	353.5	56.0	7.6	9.0	39.0	101.3	40	20W	358.5	26.5	7.5
B_3	5	359.5	42.0	48.7	14.0	38.0	128.6	40	35W	359.0	18.0	7.0
D_5	5	8.0	48.0	12.5	19.5	43.0	306.1	45	30NW	2.5	25.5	4.5
II	5	8.5	40.0	385.9	8.5	38.5	686.7	40	20W	358.5	26.0	3.0
III	4	359.5	44.0	183.1	3.0	39.5	189.1	65	25N	358.0	16.0	6.5
V	5	4.0	46.0	617.9	5.0	44.0	767.3	50	30N	352.0	20.5	3.0
IX	5	8.0	1.5	5.0	9.0	39.0	244.4	40	35W	354.0	16.0	5.0
					Pennsylvania Triassic							
							102.4			359.5	23.0	3.0

Explanation: S = number of the site; N = number of samples; D = mean declination; I = mean inclination; K = precision parameter [*Fisher*, 1953]; subscript 0 designates natural remanent magnetization; subscript 150 designates magnetization after cleaning in a peak field of 150 oe; α_{95} = radius of the circle of confidence at the 95% probability level [*Fisher*, 1953].

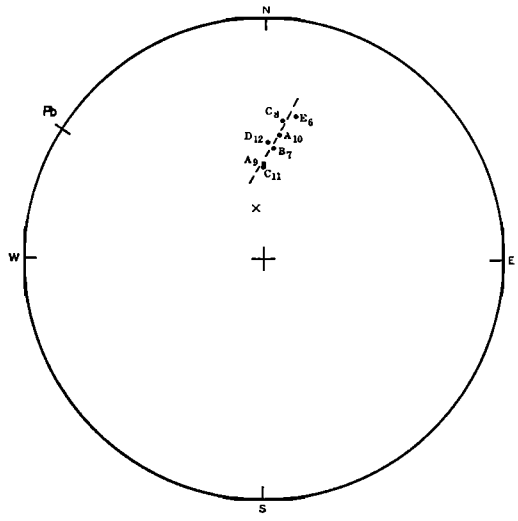
was used by *Opdyke* [1961] for his work in the Triassic diabase and basalt of New Jersey.

The paleomagnetic data for these 20 sites are summarized in Table 1. All sites have normal polarity; that is, north-seeking directions of magnetization are oriented below the horizontal.

Pole position. The mean direction of magnetization for the diabase intrusions implies a Triassic virtual geomagnetic pole at 62°N, 105°E. This direction of magnetization and other statistical parameters for the Pennsylvania diabase (Table 1) were calculated from site-mean directions by the method of *Fisher* [1953]. The semiaxes of the oval of confidence at the 95% confidence level about the virtual geomagnetic pole are 2° in the direction of the paleomagnetic colatitude and 3° at right angles to the colatitude. Equal-area projections of the site-mean directions of magnetization (after magnetic cleaning at 150 oe) are shown in Figures 2, 3, and 4. The data for each major intrusion are presented twice, demonstrating the effect of graphical 'unfolding' of dips acquired after magnetization.

Opdyke [1961] and *Cox and Doell* [1960] list a total of nine references which contain data for approximately thirty independent determinations of Triassic virtual geomagnetic poles for North America. However, most of these pole positions are unsatisfactory by present standards because (1) the directions of magnetization on which they are based may have been

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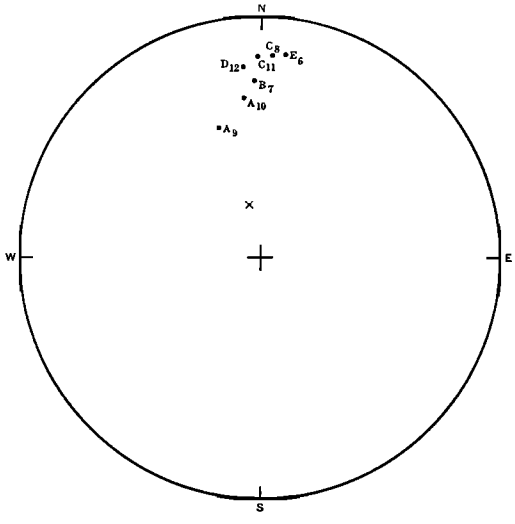
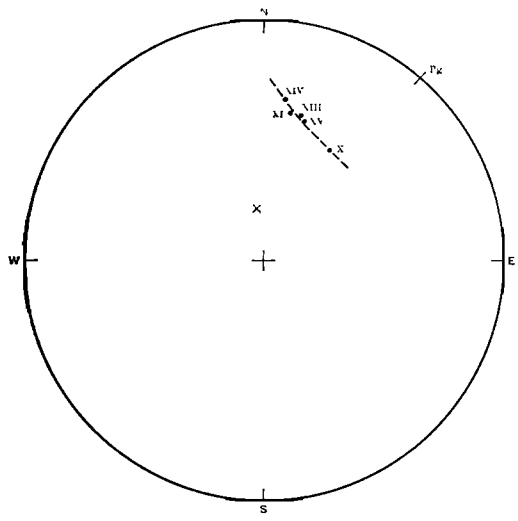


Fig. 2. Site-mean directions of magnetization for the Birdsboro diabase body before (*A*) and after (*B*) correction for attitude of adjacent layers. All specimens demagnetized in an alternating field of 150 oe. Plot is on lower hemisphere of equal-area projection. Dotted line indicates the estimated best-fitting small circle, whose pole is shown as *Pb*. *X* defines the direction of the present geomagnetic field.



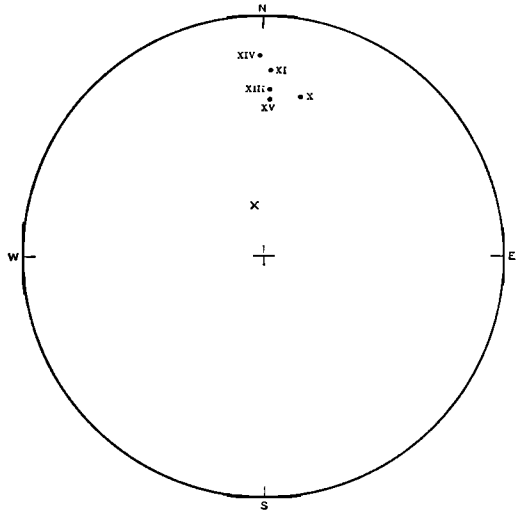


Fig. 3. Site-mean directions of magnetization for the Gettysburg diabase body before (*A*) and after (*B*) correction for attitude of adjacent strata. All specimens demagnetized in an alternating field of 150 oe. Plot is on lower hemisphere of equal-area projection. Dotted line indicates the estimated best-fitting small circle, whose pole is shown as *Pg*. *X* defines the direction of the present geomagnetic field.

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affected by significant amounts of unstable magnetization and (2) the rock samples may have acquired their magnetization in a time too short to permit the elimination of error introduced by secular variation. Six poles that appear to meet present standards are listed in Table 2. The average of these six poles, computed by the method of *Fisher* [1953], is listed at the bottom of Table 2. There is very good agreement between the two Gettysburg-Newark poles, the composite Moenkopi pole, and the Triassic average. All these virtual geomagnetic poles fall in present-day Siberia (see Figure 5).

Triassic geomagnetic secular variation. Creer [1962] proposed the existence of a quantitative relationship between paleomagnetic dispersion and paleolatitude, based on the assumption that the intensity of secular variation relative to the main geomagnetic field has remained substantially constant throughout geologic time. This dispersion consists of scatter in directions of magnetization produced in slowly magnetized rocks by secular variation in the direction of the geomagnetic field. This scatter, when measured in terms of the angular dispersion factor, σ [*Wilson,* 1959], is believed to range from a maximum of about 21° at the paleomagnetic equator to less than 10° near the corresponding pole. To test his hypothesis Creer assembled data for 18 igneous bodies, ranging in age from Permian to Quaternary and in paleolatitude from 3° to 80°, for all of which detailed paleomagnetic data are available. The results suggest a latitude dependence much like the one predicted by theory.

Dispersion in direction of magnetization for the Pennsylvania diabase is considerably less than that predicted by Creer's relationship. The observed dispersion, calculated after allowance for statistical error by the method of *Watson and Irving* [1957], is 7.5°. The paleolatitude of the diabase intrusives is 12°; the corresponding angular dispersion factor predicted by Creer's data is approximately 18°. Not only is the measured dispersion much lower than the predicted value, it also falls far below the lowest dispersion recorded by *Creer* [1962] for a comparable paleolatitude. Furthermore, the angular dispersion factor calculated for the Pennsylvania diabase is probably too high, owing to scatter introduced by the inaccuracy of dip corrections discussed below.

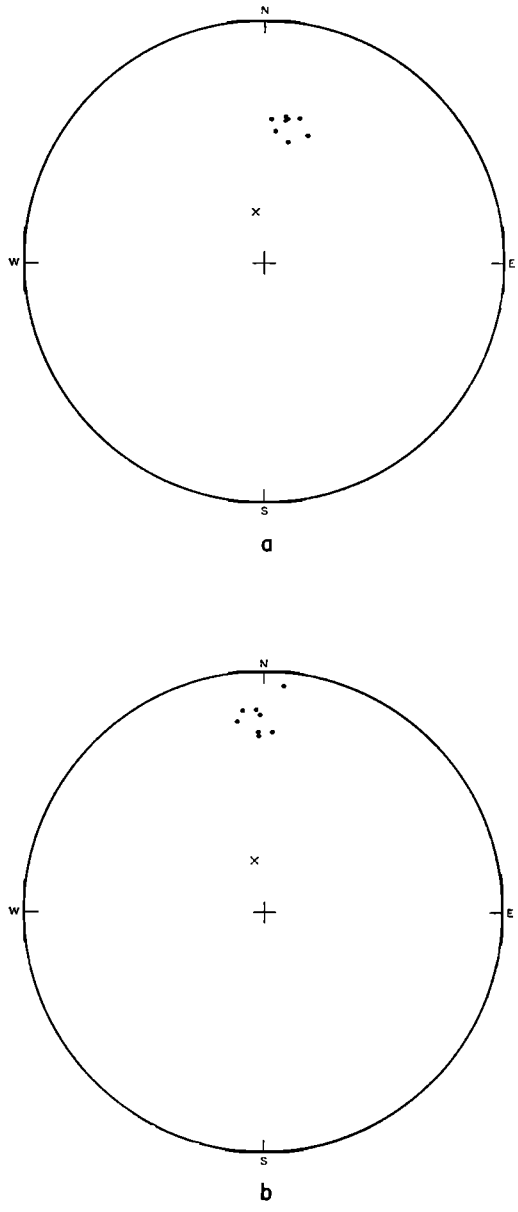


Fig. 4. Site-mean directions of magnetization for the Yorkhaven diabase body before (*A*) and after (*B*) correction for attitude of adjacent strata. All specimens demagnetized in an alternating field of 150 oe. Plot is on lower hemisphere of equal-area projection. *X* defines the direction of the present geomagnetic field.

If the time elapsed during magnetization of the Pennsylvania diabase was very short (less than several thousand years), the full range of Triassic secular variation would probably not

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TABLE 2. Triassic Virtual Geomagnetic Poles for North America

Sampling Area	Formation	Lithology	Pole Position		Reference
			Lat.	Long.	
1. Nova Scotia		Lava	77°N	75½°E	Bowker, 1959, cited by *Cox and Doell* [1960, p. 751]
2. Utah and Arizona	Moenkopi	Clastic sediments	62°N	103°E	*Runcorn* [1956], *Kintzinger* [1957], and *Collinson and Runcorn* [1960]; seven localities, summarized by *Cox and Doell* [1960, pp. 716–717]
3. Wyoming	Chugwater	Redbeds	48°N	112½°E	*Collinson and Runcorn* [1960]
4. Connecticut, Massachusetts, New Jersey	Newark group	Redbeds, basalt	55°N	88°E	*DuBois et al.* [1957] *Irving and Banks* [1961]
5. New Jersey	Newark group	Redbeds, basalt, diabase	63°N	108°E	*Opdyke* [1961]
6. Pennsylvania	Newark group	Diabase	62°N	105°E	This paper

North American Triassic

Lat.	Long.	K
62°N	101°E	51.3

have been recorded in the magnetization of the rock, and a low dispersion might have been recorded. However, the excellent agreement (Figure 5, points 2, 5, and 6) between the Pennsylvania Triassic pole and those for New Jersey [*Opdyke,* 1961] and the Moenkopi formation in Utah and Arizona [*Cox and Doell,* 1960] makes this explanation appear unlikely. Both of the latter studies involved considerable thicknesses of sediments, the times of accumulation of which probably exceeded the period of geomagnetic secular variation by several orders of magnitude.

There seems to be no a priori reason, however, to expect a latitude dependence of secular variation to have remained rigidly constant throughout geologic time. Recent data from other Upper Triassic exposures in the eastern United States seem to support the results from southeastern Pennsylvania; both the combined sediments, intrusives, and flows of New Jersey [*Opdyke,* 1961] and the lavas of Massachusetts [*Irving and Banks,* 1961] appear to have yielded angular dispersion factors at least 30% lower than those predicted by Creer's hypothesis. The low dispersion recorded in the diabase of southeastern Pennsylvania may thus represent a period of relative geomagnetic 'quiet' during the Late Triassic. Many additional studies, widely separated in present geographic position and in paleolatitude, would be necessary to establish the existence of such a period of slight secular variation and to determine whether it was of local or world-wide distribution. It is interesting to note that *Cox* [1962], in a study similar to that of *Creer* [1962], pointed out a striking modern example of an area of anomalously low geomagnetic secular variation, centered in the mid-Pacific region.

Geological implications. One interesting aspect of the equal-area plots (Figures 2, 3, and 4) is the linear distributions of site-mean directions of magnetization displayed by the Birdsboro and Gettysburg bodies (2*A*, 3*A*) before correction for geologic dip. Several explanations are available for streaked distributions of direc-

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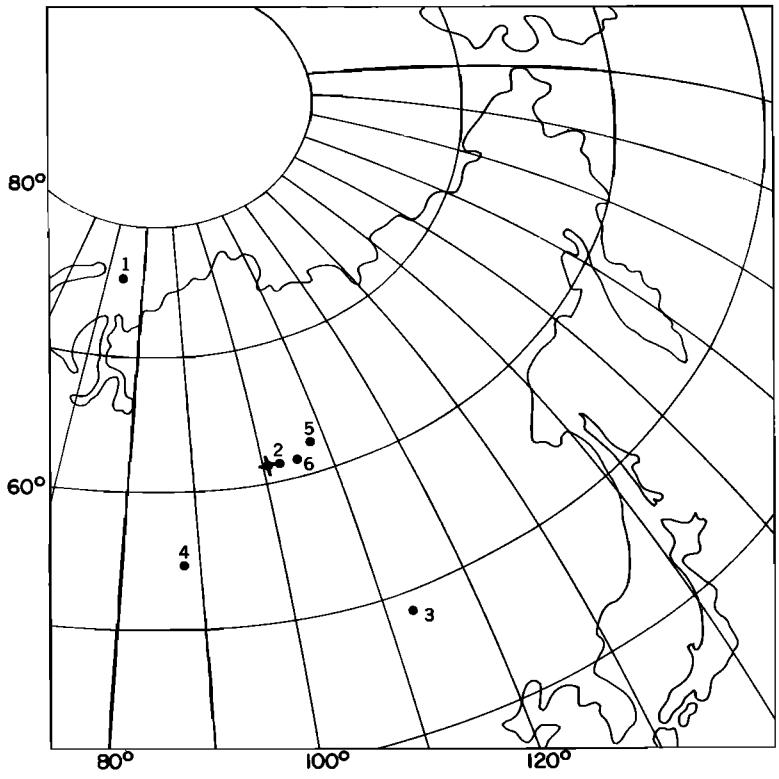


Fig. 5. Virtual geomagnetic poles for the Triassic of North America. Poles are numbered as in Table 2. The symbol + designates the average Triassic pole for North America.

tions of magnetization. That most commonly invoked involves a planar alignment developed by the vectorial addition of a viscous magnetization to a remanent magnetic moment. Such streaking has as end points the true direction of remanence and the direction of the ambient field at the time the viscous moment was acquired, presumably that of the present geomagnetic field. Such an explanation could account for the observed linear distribution of the Birdsboro diabase body (Figure 2*A*), but it is probably inappropriate in view of the following observations. (1) Alternating-field demagnetization should eliminate any viscous magnetization originally present. On first measurement, 5 of the 20 sites yielded linear distributions of a type attributable to viscous magnetization, but this streaking was destroyed by the demagnetization process. None of the sites shows streaking after demagnetization at a peak field of 150 oe. (2) Streaking due to viscous magnetization is most conspicuous when many observations from a single igneous body or formation are plotted. The process of calculating site means would tend to eliminate this feature. (3) This process does not explain the linear distribution displayed by the Gettysburg plot (Figure 3*A*), which has an entirely different trend.

Variations in the strength of magnetic anisotropy [*Khan,* 1962; *Uyeda et al.,* 1963; *Hargraves,* 1959; *Strangway,* 1961] might produce linear distributions of directions of magnetization. Such streaking would be apparent in a plot of mean directions from widely scattered sites and would not be reduced by magnetic cleaning. However, there is no geological reason for suspecting strong anisotropy in the Pennsylvania diabase; published petrographic descriptions offer no hint of the existence of the particular mineralogy or texture required. Also, in theory, most instances of streaking attributable to magnetic anisotropy in tabular bodies should be confined to a plane containing the direction of the paleomagnetic field and oriented at right

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angles to the plane of the tabular rock mass. Neither the Birdsboro nor the Gettysburg diabase body displays streaking oriented in this manner.

Linear distribution of site-mean directions of magnetization also can result from differential tectonic rotation. Differential rotation of a magnetized rock mass would result in the distribution of its directions of magnetization along a small circle in equal-area projection. The pole of this small circle would be the axis of folding, or rotation, of the magnetized body. Figures 2*A* and 3*A* show what may be such small-circle distributions; furthermore, the poles of these small circles lie within a few degrees of the regional strike of the two bodies in question. As is apparent from Figure 1, the Gettysburg diabase body, although complicated by major offshoots, changes of thickness, and other irregularities, maintains a grossly linear, concordant aspect throughout most of its length. Its regional strike (42) falls essentially on the pole of the small circle defined by site-mean directions of magnetization for the Gettysburg body (41). There is also excellent agreement between the pole of the streaked Birdsboro plot (292) and the apparent strike of the segment of the intrusion from which most of these samples were taken (297). According to this explanation, which seems to account most readily for the evidence now available, the Yorkhaven diabase body does not show such a linear distribution for one of two reasons; either it lacks significant changes in dip or, if these are present, their effect is masked by the lack of linearity of the intrusion.

Minimum differential rotations implied by these distributions of directions of magnetization are 30° for the Gettysburg body and 20° for the Birdsboro body. It is interesting to note that, owing to the scattered, slumped, and, in places, grossly distorted condition of available exposures of sedimentary rock in the neighborhood of the diabase bodies, the existence of these postulated changes in dip would not be obvious during routine geologic mapping. The possibility that adjacent sedimentary attitudes do not adequately describe the structure of the diabase is suggested by the fact that some linearity remains after correction for the best available geologic dip (Figures 2*B* and 3*B*). The tight grouping of Upper Triassic virtual geomagnetic poles shown in Figure 5 indicates that these inaccuracies in dip correction have introduced little error into the calculation of a mean geomagnetic field direction for the Upper Triassic in Pennsylvania. Evidently the errors are random, leaving the mean direction of magnetization unchanged (but increasing the dispersion).

If additional work verifies the hypothesis that these linear distributions are the result of differential tectonic rotations, paleomagnetism will have been established as a potential structural tool for the Triassic of southeastern Pennsylvania. The validity of the observation, suggested by Figures 1, 2*A*, and 3*A* and Table 1, that on the average those sites having the steepest dips, as determined by their directions of magnetization, are nearest the northern border of the Triassic basin will be tested with additional data.

If the Gettysburg and Birdsboro diabase bodies acquired their magnetization at approximately the same time, and did so before any significant amount of Triassic deformation had taken place, and if the only structural displacements they have undergone are variable amounts of rotation about their respective lines of strike, small-circle distributions of directions of magnetization much like those of Figure 2*A* and 3*A* would be the result. However, the actual sense of Triassic deformation was such as to develop northerly dips throughout the basin, thus steepening the inclination of remanent magnetic moments. Small-circle distributions attributable solely to deformation of this sort would not intersect. Instead, each small-circle distribution, representing the product of differential steepening around local lines of strike, would tend to diverge from the common, original direction, the amount of divergence in each instance being equal to the smallest amount of tectonic rotation experienced by any sampling site.

The actual case is somewhat different, as can be observed by a close examination of plots 2*A* and 3*A*. The streaked distribution of site-mean directions for the Gettysburg and Birdsboro diabase bodies intersect, and the direction thus defined (declination 9.5°, inclination +33.0°) is markedly different from the average direction for the Pennsylvania Triassic (declination 359.5°, inclination +23.0°). To derive the latter direction for the former requires a rotation of slightly

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more than 20° about a northeast-trending axis, the sense of rotation being such as to develop dips toward the northwest. Rotations of this direction and magnitude are substantially those necessary to develop the monoclinal dips found in the Triassic sedimentary rocks throughout southeastern Pennsylvania. Thus the deformation of the Birdsboro and Gettysburg diabase bodies can probably be explained by two processes: differential rotation of 20° to 30° around their local lines of strike and a superimposed general northwestward tilt. It is perhaps significant that for the Gettysburg diabase body, the northeast strike of which is roughly parallel to the axis of this presumed regional deformation, the differential rotations included both shallowing and steepening of dip, whereas for the Birdsboro body, which strikes nearly at right angles to the regional trend, the differential rotations all took the form of increases in the angle of dip.

Several other conclusions concerning the structural history of the Gettysburg-Newark basin can be drawn from the remanent magnetic data. The excellent agreement in direction of magnetization between sedimentary rocks and intrusive igneous rocks in New Jersey [*Opdyke*, 1961] and Pennsylvania indicate that very little dip had been acquired by the sedimentary rocks before intrusion. After correction for present geologic dip, these three directions agree to within about 5°, making it appear unlikely that dips much in excess of this figure were present in the sedimentary rocks at the time of diabase intrusion (as suggested, for instance, by *Stose* [1932, p. 54]). The sedimentary rocks appear to have been nearly horizontal at the time of the basaltic igneous activity, much as required by *Sanders'* [1963] graben-subsidence hypothesis.

Finally, two minor items concerning *Sanders'* [1963] proposed structural history and the hypothesis advanced by *Drake and Woodward* [1963] to explain the curvature of Appalachian structural trends should be noted. No evidence has been found of a stable direction of magnetization aligned at a large angle to the predominant Triassic direction, as would be expected if a second, postdeformational generation of diabase intrusions were present. This is not conclusive, however, since most of the sampling has been confined to the large diabase bodies which are representatives of Sanders' first generation of intrusions. Likewise, the rotations of directions of magnetization which would accompany bending of the Triassic basin are not present. Rotations of the sort required by the proposal of Drake and Woodward would produce deflections of as high as 50° in the corrected magnetic declination of any rock unit involved, leaving the inclination essentially unaltered. The fact that such discrepancies are not observed, however, merely indicates that the proposed bending would have to have been pre-Triassic, a probability fully recognized by *Drake and Woodward* [1963, p. 59].

Magnetic properties. In Table 3 the available data on magnetic properties are summarized. No petrographic studies have been made of these samples; attempts to correlate magnetic properties with lithology are limited to gross petrologic variations apparent in hand specimens. Three groups have been distinguished, largely on the basis of grain size, and average magnetic parameters calculated for each group. The statistical significance of these figures is questionable, owing to the small number of samples involved; nevertheless, Table 3 contains several items of potential significance to the study of rock magnetism and exploration geophysics.

Examination of these preliminary data suggests a correlation between high total magnetization and the coarser-grained diabase facies, having lenses and dikes of very coarse grained, relatively leucocratic hornblende-bearing material which probably formed from late magnetic fluids. Both magnetic susceptibility and remanent magnetism for diabase of this type are markedly higher than for finer-grained material, probably owing to the concentration of magnetite in the coarse diabase. There also seems to be some evidence that the thermoremanent magnetization of this late-stage material is the least stable of the lot. This is suggested by the ratio log K_{150}/K_0, large values of which indicate a significant reduction of scatter by the demagnetization process and hence the presence of a considerable component of unstable magnetization before magnetic cleaning.

Another interesting relationship suggested by Table 3 is the correlation of low values of log K_{150}, indicating relatively high scatter, with sites located in fine-grained to aphanitic diabase. This could reflect greater magnetization time,

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TABLE 3. Magnetic Properties and Density of Triassic Diabase from Southeastern Pennsylvania

Sample	$J_0 \times 10^4$		$J_{150} \times 10^4$		$k \times 10^3$		Log K_{150}/K_0		Log K_{150}		Specific gravity	
	Range	Average	Range	Average	Range	Average	Range	Average	Range	Average	Range	Average
Coarse-grained granophyric diabase (20 samples)	2.06 to 20.60	11.07	1.76 to 16.81	8.29	0.18 to 2.88	1.75	0.251 to 1.431	1.072	1.477 to 2.837	2.195	2.95 to 3.07	3.00
Medium-grained 'normal' diabase, with little granophyre (52 samples)	1.59 to 52.38	8.29	1.44 to 3.66	2.61	0.13 to 2.12	0.77	−0.013 to 1.424	0.736	1.730 to 3.018	2.341	2.92 to 3.10	3.02
Aphanitic to fine-grained diabase, comprising small dikes or contact facies of large bodies (23 samples)	3.48 to 15.32	7.09	1.57 to 14.45	4.81	0.37 to 1.30	0.82	0.000 to 1.694	0.686	1.454 to 2.388	1.884	2.97 to 3.02	3.00

Explanation: J = intensity of magnetization in cgs emu/cc; k = magnetic susceptibility in cgs emu/cc; K = precision parameter [*Fisher*, 1953]; subscript 0 designates natural remanent magnetization; subscript 150 designates magnetization after cleaning in a peak field of 150 oe.

as determined by the velocity of the Curie point isotherm, for the outer parts of large igneous bodies [*Jaeger*, 1957], thus producing more scatter from secular variation. However, of the five sites classified as aphanitic to fine-grained, two (numbers X and XV) are located on thin dikes, and they presumably acquired their magnetization in a relatively short period of time. Considerably more data are needed before any detailed correlation between lithology, cooling history, and magnetic properties can be advanced.

One magnetic parameter of especial interest in exploration geophysics is the Q factor, defined [*Koenigsburger*, 1938] as the ratio of remanent to induced magnetization. Large values of Q indicate that the dominant magnetization of the body in question is remanent, and therefore may have a direction (or polarity) widely at variance with that of the earth's present magnetic field. Q for the Pennsylvania diabase ranges from 0.26 to 7.62; the average is 1.89. However, the remanent magnetization used in calculating Q for this suite of rocks in many cases includes a considerable component of 'soft' magnetization, much of which seems to be nearly random in orientation. Some part of such random moments might be expected to cancel out, thus reducing the effective contribution of remanent magnetization to the magnetization measured by an airborne magnetometer. Preliminary calculation suggested that the net effect of this cancellation of random magnetic moments is to lower the effective value of Q by about 10%. For the Triassic diabase of Pennsylvania, in its present structural position, remanent and induced magnetization are additive; however, if the polarity of these intrusions were reversed, the preliminary data on magnetic properties indicate that the intrusions would appear nearly nonmagnetic to an airborne magnetometer.

Acknowledgments. I wish to thank Allan Cox, Andrew Griscom, and Harry Smedes for their help as reviewers. William Huff and Daniel Stinnett assisted in the laboratory and Peter Shive in the field.

Publication authorized by the Director, U. S. Geological Survey.

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(Manuscript received February 20, 1965.)

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